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United States Patent	12411887
Kind Code	B2
Date of Patent	September 09, 2025
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Transformer-based object detection

Abstract

Object detection using a transformer-based object detection model includes randomly initializing basic queries for the model, modulating the basic queries based on semantics of input images, and training the model basic on features extracted from input images and the modulated queries.

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Appl. No.: 18/057008

Filed: November 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240168991 A1	May. 23, 2024

Publication Classification

Int. Cl.: G06V10/77 (20220101); G06F16/538 (20190101); G06V10/42 (20220101); G06V10/764 (20220101); G06V10/774 (20220101); G06V10/82 (20220101); G06V20/70 (20220101)

U.S. Cl.:

CPC G06F16/538 (20190101); G06V10/42 (20220101); G06V10/7715 (20220101); G06V10/774 (20220101); G06V10/82 (20220101); G06V20/70 (20220101);

Field of Classification Search

CPC: G06F (16/538); G06F (16/532); G06F (16/53); G06F (16/583); G06V (10/42); G06V (10/7715); G06V (10/774); G06V (10/82); G06V (20/70)

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Background/Summary

TECHNICAL FIELD

(1) The embodiments described herein pertain generally to object detection within the context of computer vision.

BACKGROUND

(2) Under the broad technology of computer vision are the localizing and categorizing one or more objects of interest in a given image. Fundamental to such tasks is object detection and object segmentation. Transformer-based detection and segmentation models use basic queries for different objects, implementing a multi-stage attention process. Thus, the fixed queries serve as global priors for both location and semantics of target objects within an input image.

SUMMARY

(3) In one example embodiment, a method of detecting objects using a transformer-based object detection model includes generating modulated queries based on basic queries of the transformer-based object detection model, training the model using the basic queries and the modulated queries, replacing the basic queries with the modulated queries as input for a transformer decoder that is to execute the trained transformer-based object detection model, and inputting the modulated queries

and extracted image features to the transformer decoder.

(4) In accordance with at least one other example embodiment, a non-volatile computer-readable medium that, when executed, causes at least one processor to perform operations related to object detection, that include: receiving an input image, generating dynamic detection queries based on basic queries and semantics of the input image, training a transformer-based object detection model using the basic queries and the dynamic detection queries, and performing object detection based on the dynamic detection queries using the trained transformer-based object detection model.

(5) In accordance with at least one other example embodiment, a transformer-based object detector includes: a feature extraction module, a global pooling module to calculate combination coefficients using output from the feature extraction module, a dynamic detection query generator to generate dynamic detection queries as a function of the combination coefficients and basic queries for a transformer-based object detection model, and a transformer decoder to receive the output from the feature extraction module and the dynamic detection queries as input for performing object detection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the detailed description that follows, embodiments are described as illustrations only since various changes and modifications will become apparent to those skilled in the art from the following detailed description. The use of the same reference numbers in different figures indicates similar or identical items.

(2) FIG. 1 shows a system in which transformer-based object detection may be implemented, arranged in accordance with at least some embodiments described and recited herein;

(3) FIG. 2A shows at least a portion of an example processing flow for implementation of transformer-based object detection, pertaining to generation of dynamic or modulated queries, in accordance with at least some embodiments described and recited herein;

(4) FIG. 2B shows at least a portion of an example processing flow for implementation of transformer-based object detection, pertaining to training of a detection model, in accordance with at least some embodiments described and recited herein;

(5) FIG. 2C shows at least a portion of an example processing flow for implementation of transformer-based object detection, pertaining to execution of the trained model, in accordance with at least some embodiments described and recited herein;

(6) FIG. 3 shows an illustrative computing embodiment, in which any of the processes and sub-processes of procuring 3D images based on text input may be implemented as executable instructions stored on a non-volatile computer-readable medium.

DETAILED DESCRIPTION

(7) In the following detailed description, reference is made to the accompanying drawings, which form a part of the description. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Furthermore, unless otherwise noted, the description of each successive drawing may reference features from one or more of the previous drawings to provide clearer context and a substantive explanation of the current example embodiment. Still, the example embodiments described in the detailed description, drawings, and claims are not intended to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described and recited herein, as well as illustrated in the drawings, may be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

(8) Additionally, portions of the present disclosure may be described herein in terms of functional block components and various processing steps. It should be appreciated that such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions.

(9) In the present description and recitation, the following terms may provide context for embodiments described, recited, and/or suggested herein; and may also be used, in addition to their accepted meaning, as follows.

(10) Artificial intelligence, alternatively referenced herein as “AI,” may refer to a learned or trained computer or processor-related technology by which decisions and/or actions are autonomously made, in place of human intervention. AI refers to software, i.e., algorithms and/or programs, hardware or firmware, or any combination thereof that supports machine learning, natural language understanding, natural language processing, speech recognition, computer vision, etc. Also included among the range of AI functions and capabilities, and pertinent to the embodiments disclosed, recited, and suggested herein, object detection or segmentation relative to an input image, images, or video.

(11) An engine or generator, as disclosed, recited, and/or suggested herein, may refer to a type of software, firmware, hardware, or any combination thereof, that facilitates generation of source code or markup to produce elements that begin another process. In addition, or alternatively, an engine or generator may facilitate automated processes, in which various software elements interact to produce an intended product, whether physical or virtual based on natural language descriptions, inputs, or other prompts. In accordance with known AI technologies, the AI engines or generators disclosed, recited, and/or suggested herein are trained in accordance with either unimodal or multimodal training models.

(12) Object detection, in accordance with computer vision and image processing, may refer to technologies that are designed, programmed, or otherwise configured to detect instances of semantic objects of a certain class in digital images and/or videos. Non-limiting contextual applications for object detection may include image retrieval, video surveillance, device security, etc.

(13) Object segmentation, in accordance with computer vision and image processing, may refer to technologies that are designed, programmed, or otherwise configured to partition a digital image into multiple image segments, i.e., image regions or image objects, to simplify and/or change a representation of the original image into a format that is more easily managed and/or analyzed.

(14) Queries, as disclosed and recited herein in accordance with known object detection and segmentation technologies, may respectively refer to a representation of an object. Each query may represent an object and having multiple representations referring to classification, box regression and mask encoding simultaneously in vector form. For known object detection and segmentation technologies, basic queries are fixed for different objects in every input image.

(15) A social media application, as disclosed and recited herein, may refer to an on-line application that allows account-holding users to interact with one another using various media and on varying scales, with such interaction including creating and/or sharing media content. As disclosed and recited herein, a user device may have an instance of social media application account stored locally or may access the user's account via a web-based version of the particular social media application.

(16) A platform, e.g., a social media platform, as disclosed and recited herein, may refer to an application on which algorithms and/or programs enabling execution or implementation of a collection of communication-based or media-sharing technologies may be hosted. Further, any algorithm or program described, recited, or suggested herein may be executed by one or more processors hosted on such a platform. Non-limiting examples of such technologies may include the creation, sharing, and/or storage of multi-media offerings.

(17) Media, or multi-media, offerings or experiences, as referenced herein, may include but not be

limited to recorded or live transmittable content including text, audio, images, animations, video, etc. In addition, such offerings or experiences may include, but again not be limited to, interactive augmented reality (AR) and/or interactive virtual reality (VR) experiences.

(18) FIG. 1 shows a system in which transformer-based object detection may be implemented, arranged in accordance with at least some embodiments described and recited herein. As depicted, system **100** includes input device **102** and object detection model **105**. Object detection model **105** includes, at least, encoder **110**, which includes feature extractor **115** and global pooling module **120**; combination coefficient generator **125**, which implements multilayer perceptron (MLP) **130**; basic queries **135**; dynamic/modulated query generator **140**; and transformer decoder **145**.

Although illustrated as discrete components, various components may be divided into additional components, combined into fewer components, or eliminated altogether while being contemplated within the scope of the disclosed subject matter. It will be understood by those skilled in the art that each function and/or operation of the components may be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or any combination thereof.

(19) Input device **102** may refer to one or more embodiments of a classical computing environment that may be, or include, a classical computer, processing device, a microprocessor, a microcontroller, a digital signal processor, or any combination thereof. Device **102** may be one of various electronic devices, or a combination thereof, having one or more image and/or video capturing components, i.e., camera and/or video recorder, display screens with audio and/or video inputs/outputs and that support the providing of and consumption of content relative to a media platform. The various electronic devices may include but not be limited to a security/surveillance device, smartphone, a tablet computer, a laptop computer, a desktop computer, an e-book reader, an MP3 (moving picture experts group audio layer III) player, an MP4, and/or any other suitable electronic devices. Non-limiting examples of input device **102** as a security device may include a video doorbell, a vehicle dash-cam, a security camera (whether constantly active or motion-activated), etc. Additional non-limiting examples of input device **102** may include a database, local server, cloud-based service, a virtual reality (VR) and/or augmented reality (AR) servers, etc. Further, any algorithm or program described, recited, or suggested herein may be executed by one or more processors hosted on input device **102**.

(20) Input **104**, in accordance with at least some of the embodiments disclosed and recited herein, may refer to digital images, digital video, text, and/or audio that may be input manually or in an automated manner to an appropriate input interface. Input **104** may be transmitted or otherwise communicated from input device **102** to a receiving component corresponding to object detection model **105** via a wired or wireless network. Such network may be regarded as a medium that is provided as a bidirectional communications link between a media platform on which object detection model **105** is hosted and input device **102**. The network may include the Internet, a local area network (LAN), a wide area network (WAN), a local interconnect network (LIN), a localized cloud, etc.

(21) Object detection model **105** may refer to AI-based implementation of, e.g., algorithms and/or programs, hardware or firmware, or any combination thereof, to categorize, detect, isolate, and/or localize objects and/or segments of interest in an input **104** image, images, and/or video. A non-limiting example of such model may be a Detection Transformer (DETR) model. Object detection model **105** may be hosted on one or multiple servers, on the order of hundreds to thousands, which may be hosted on a cloud-based infrastructure. Further, object detection model **105** may be implemented by single or multiple classical computers, and facilitate transmissions with one or more of input devices **102** across single or multiple connections or channels.

(22) As described and recited herein, object detection model **105** may be regarded as a transformer-based encoder-decoder architecture, e.g., DETR. Accordingly, an encoder includes encoding layers that processes input iteratively and a decoder process output iteratively.

(23) Encoder **110** may refer to a component or module that is designed, programmed, or otherwise

configured to receive input **104** image, images, and/or video from input device **102**. For the purposes of this description, but not in any limiting manner, encoder **110** may include feature extractor **115** and global pooling module **120**.

(24) Feature extractor **115** may refer to a component or module that is designed, programmed, or otherwise configured to iteratively extract, from input **104** received at encoder **110**, information from the input **104** content in accordance with known extraction technologies. Non-limiting examples of extracted features may include persons (intact or in part), objects, edges, points, boundaries, curves, shapes, etc. Such features may be regarded as high-level content of the input images, typically corresponding to semantics of the respective images.

(25) As indicated prior, non-limiting contextual applications for object detection may include image editing, video surveillance, device security, etc., and therefore a definition or parameters of a feature that is subject to extraction by feature extractor **115** may be dependent upon the contextual application.

(26) Global pooling module **120** may refer to a component or module corresponding to object detection model **105** that is designed, programmed, or otherwise configured to determine a global average pooling of the extracted feature vectors or maps corresponding to features corresponding to input **104** that are extracted by feature extractor **115**.

(27) Combination coefficient generator **125** may refer to a component or module corresponding to object detection model **105** that is designed, programmed, or otherwise configured to generate a combination coefficient by inputting the calculated global feature, which includes semantics of input **104**, to MLP **130**.

(28) MLP **130** refers to a multilayer perceptron (MLP), which is a feedforward artificial neural network, designed, programmed, or otherwise configured to generate combination coefficients, in accordance with the embodiments described, recited, and suggested herein. For the purposes of this description, but not in any limiting manner, MLP **130** may affect a global average pooling of the feature vectors or maps determined by global pooling module **120** corresponding to encoder **110** to generate combination coefficients corresponding to semantics of input **104**, serving as a basis for dynamic/modulated queries.

(29) Basic queries **135** may refer to fixed queries corresponding to object detection model **105** that each represent an object and having multiple representations for class, box, mask, etc., for the transformer-based object detection model **105**. Basic queries **135** are each a fixed vector and are therefore the same for any image input to object detection model **105**.

(30) Dynamic/modulated query generator **140** may refer to a component or module corresponding to object detection model **105** that is designed, programmed, or otherwise configured to generate dynamic/modulated queries by modulating the initialized basic queries. Dynamic/modulated query generator **140** performs a convex combination on the generated combination coefficients and the initialized basic queries. Thus, because the combination coefficients are generated based on the global features of the feature maps of input **104** images, dynamic/modulated queries are generated in accordance with the semantics for each input **104** image.

(31) Transformer decoder **145**, as described, recited, and suggested herein, may refer to a trained executing component or module for object detection model **105**. In accordance with the embodiments described and recited herein, transformer decoder **145** is designed, programmed, or otherwise configured to detect, as trained, objects in accordance with the semantics of the input **104** images.

(32) As shown in FIG. 1, transformer decoder **145** receives input for training purposes that includes the initialized basic queries and the generated dynamic/modulated queries. The dynamic/modulated queries are modulations of the basic queries based on the combination coefficients that are generated based on the global features of the feature maps of input **104** images.

(33) Transformer decoder **145** is further designed, programmed, or otherwise configured to generate or produce the output **150** of transformer-based object detection model **105**. In accordance

with the embodiments described and recited herein, trained transformer decoder **145** receives features extracted from input **104** from feature extractor **115** and also receives dynamic/modulated queries from dynamic/modulated query generator **140**. That is, the initialized basic queries are input to transformer decoder **145** for training purposes only but are not input to transformer decoder **145** for execution or implementation of object detection. Thus, the input to transformer decoder **145** includes the extracted features and the dynamic/modulated queries.

(34) Object detection model **105**, including transformer decoder **145**, may refer to an AI algorithm, program, component, and/or module that is hosted on or remotely and communicatively corresponds to a media platform, including but not limited to a social media platform, a security/surveillance platform, etc. Transformer decoder **145** may be designed, programmed, or otherwise configured to output **150** classifications, boxes, and/or masks corresponding to objects within input **104** images that are input or subject to objection detection model **105**.

(35) Object detection model **105** may be implemented in accordance with an on-line application that allows account-holding users to interact with one another using various media and on varying scales, with such interaction including creating and/or sharing media content. As disclosed and recited herein, device **102** may have an instance of social media application account stored locally or may access a user's account via a web-based version of the particular social media application.

(36) A platform, e.g., a social media platform and/or a security/surveillance platform, as disclosed and recited herein, may refer to an application on which algorithms and/or programs enabling execution or implementation of a collection of communication-based or media-sharing technologies may be hosted. Such technologies include monitoring, creating, sharing, and/or storing multi-media offerings.

(37) Any one or more of a server or cluster of servers upon which a media platform is hosted, and object detection model **105** is executed or implemented, may refer to a high-performance computing (HPC) environment that includes, at least, a CPU and GPU that is present on a, e.g., a video card, embedded on a motherboard, or on the CPU die. The training and/or resulting object detection may be executed entirely on the CPU or in part on the CPU and the GPU. Alternative embodiments may be executed in evolved HPC components known in the art. Regardless, the CPU, GPU, and/or HPC components may store one or more algorithms and/or programs that, when executed thereon, may cause the execution or performance of operations and/or functionality as disclosed and/or recited herein. Also, a computer-readable non-volatile medium may be provided according to the embodiments described herein. The computer readable medium stores computer programs. The computer programs are used to, when being executed by a processor, execute or perform the operations or functionality in connection with at least the embodiments described and recited herein.

(38) FIGS. 2A-2C may be combined to show an example processing flow for implementation of transformer-based object detection and/or segmentation, in accordance with at least some embodiments described, recited, and even suggested herein. It is understood that each function and/or operation shown and described with reference to any of FIGS. 2A-2C may be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or any combination thereof, such as those illustrated, described, and even suggested in accordance with FIG. 1.

(39) FIG. 2A shows example sub-processing flow **200A** for implementation of transformer-based object detection, pertaining to generation of dynamic/modulated queries, in accordance with at least some embodiments described and recited herein. As shown, generation of dynamic/modulated queries includes at least blocks **205**, **210**, **215**, **220**, and **225**.

(40) FIG. 2B shows example sub-processing flow **200B** for implementation of transformer-based object detection, pertaining to training of a detection model, in accordance with at least some embodiments described and recited herein. As shown, training of a detection model includes at least blocks **220**, **225**, and **230**.

(41) FIG. 2C shows example processing flow **200C** for training and implementing transformer-based object detection, pertaining to execution of the trained model, in accordance with at least some embodiments described and recited herein. As shown, the detecting of objects includes at least blocks **230**, **235**, **240**, and **245**.

(42) Illustrated sub-processes **200A**, **200B**, and **200C** may be combined for transformer-based object detection and/or segmentation. Further, the description of **200B** may overlap with that of **200A** and the description of **200C** may overlap with that of **200B**. Thus, it is to be understood that there is no clear distinction between **200A**, **200B**, and **200C**, which are divided herein for illustrative and descriptive purposes only; and the descriptions are not intended to be so limiting in any manner.

(43) Process **200A** for generating dynamic or modulated queries may begin at block **205**.

(44) At block **205** (receive input image), encoder **110** may receive input **104** image, images, and/or video from input device **102**, which may be one of various electronic devices, or a combination thereof, having one or more image and/or video capturing components, i.e., camera and/or video recorder, display screens with audio and/or video inputs/outputs and that support the providing of and consumption of content relative to a media platform. Non-limiting examples of device **102** may include a smartphone, a tablet computer, a laptop computer, a desktop computer, etc.; therefore, input **104** may be received as an image, series of continuous images, and/or video. Processing may proceed to block **210**.

(45) At block **210** (extract image feature(s)), feature extractor **115** extracts, from input **104** received at encoder **110**, information from input **104** using known extraction techniques by which high-level content, i.e., semantics, of input **104** may be extracted. Processing may proceed to block **215**.

(46) At block **215** (generate global feature), global pooling module **120** determine a global average pooling of the extracted feature vectors or maps that respectively correspond to the high-level content, i.e., semantics, of content **104** that is extracted by feature extractor **115**. Processing may proceed to block **220**.

(47) At block **220** (generate combination coefficients), MLP **130** generates a combination coefficient by inputting the determined global feature to combination coefficient generator **125**. Processing may proceed to block **230**.

(48) At block **225** (generate dynamic/modulated queries), dynamic/modulated query generator **140** converts initialized basic queries **135** into dynamic/modulated queries by performing convex combination of the generated combination coefficients and the initialized basic queries.

(49) Processing may proceed to the description of sub-processing flow **200B**, pertaining to training of a detection model, which may include the operation at block **220**.

(50) At block **220** (input of global feature to MLP), combination coefficients are generated by inputting a generated global feature to MLP **130**, corresponding to combination coefficient generator **125**. Processing may proceed to block **225**.

(51) At block **225** (multiply convex combination and basic queries . . .), dynamic/modulated queries are generated by a convex multiplication between the combination coefficients generated by MLP **130** and the initialized basic queries for the transformer-based object detection model at dynamic/modulated query generator **140**. Processing may proceed to block **230**.

(52) At block **230** (input basic queries and dynamic/modulated queries . . .), transformer decoder **145** is trained for subsequent predictions for object detection based on the initialized basic queries for the transformer-based object detection model and the generated dynamic-modulated queries that are based on the semantics of input **104** image or images.

(53) Processing may proceed to the description of sub-processing flow **200C**, pertaining to training and detecting objects in accordance with the trained transformer-based object detection model **105**, which may include the operation at block **230**.

(54) At block **230** (input basic queries and dynamic/modulated queries to model), initialized basic queries and generated dynamic/modulated queries are input to train transformer decoder **145**.

Processing may proceed to block **235**.

(55) At block **235** (train model), transformer decoder **145** is trained based on the input from block **230**, including basic queries corresponding to the model and modulations of the basic queries. Accordingly, it is understood that hundreds or even thousands of iterations of operations **205-230** may be repeated to train the transformer based object detection model.

(56) At block **240** (input extracted features and dynamic/modulated queries . . .), transformer decoder **145** executes the trained model based on the input of the dynamic/modulated queries generated by dynamic/modulated query generator **140** and extracted features from feature extractor **115**. That is, in accordance with the embodiments described, recited, and suggested herein, while the initialized basic queries are input to transformer decoder **145** for training purposes, the basic queries are not input to transformer decoder **145** for execution or implementation of object detection. Processing may proceed to block **245**.

(57) At block **245** (output class & box/mask), transformer decoder **145** executes object detection in accordance with the trained model by outputting classifications of boxed and/or masked objects in an input image.

(58) FIG. **3** shows an illustrative computing embodiment, in which any of the processes and sub-processes transformer-based object detection may be implemented as executable instructions stored on a non-volatile computer-readable medium. The computer-readable instructions may, for example, be executed by a processor of a device, as referenced herein, having a network element and/or any other device corresponding thereto, particularly as applicable to the applications and/or programs described above corresponding to system **100**.

(59) In a very basic configuration, a computing device **300** may typically include, at least, one or more processors **302**, a memory **304**, one or more input components **306**, one or more output components **308**, a display component **310**, a computer-readable medium **312**, and a transceiver **314**.

(60) Processor **302** may refer to, e.g., a microprocessor, a microcontroller, a digital signal processor, or any combination thereof.

(61) Memory **304** may refer to, e.g., a volatile memory, non-volatile memory, or any combination thereof. Memory **304** may store, therein, an operating system, one or more applications corresponding to model **105** and/or program data therefore. That is, memory **304** may store executable instructions to implement any of the functions or operations described above and, therefore, memory **304** may be regarded as a computer-readable medium.

(62) Input component **306** may refer to a built-in or communicatively coupled keyboard, touch screen, or telecommunication device. Alternatively, input component **306** may include a microphone that is configured, in cooperation with a voice-recognition program that may be stored in memory **304**, to receive voice commands from a user of computing device **300**. Further, input component **306**, if not built-in to computing device **300**, may be communicatively coupled thereto via short-range communication protocols including, but not limitation, radio frequency or Bluetooth®.

(63) Output component **308** may refer to a component or module, built-in or removable from computing device **300**, that is configured to output commands and data to an external device.

(64) Display component **310** may refer to, e.g., a solid state display that may have touch input capabilities. That is, display component **310** may include capabilities that may be shared with or replace those of input component **306**.

(65) Computer-readable medium **312** may refer to a separable machine-readable medium that is configured to store one or more programs that embody any of the functions or operations described above. That is, computer-readable medium **312**, which may be received into or otherwise connected to a drive component of computing device **300**, may store executable instructions to implement any of the functions or operations described above. These instructions may be complimentary or otherwise independent of those stored by memory **304**.

(66) Transceiver **314** may refer to a network communication link for computing device **300**, configured as a wired network or direct-wired connection. Alternatively, transceiver **314** may be configured as a wireless connection, e.g., radio frequency (RF), infrared, Bluetooth®, and other wireless protocols.

(67) From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Aspects

(68) Aspect 1. A method of detecting objects using a transformer-based object detection model, comprising: generating modulated queries based on basic queries of a transformer-based object detection model; training the transformer-based object detection model using the basic queries and the modulated queries; replacing the basic queries with the modulated queries as input for a transformer decoder that is to execute the trained transformer-based object detection model; and inputting the modulated queries and extracted image features to the transformer decoder.

(69) Aspect 2. The method of Aspect 1, wherein the generating of the modulated queries includes generating convex combinations of the basic queries to produce the modulated queries.

(70) Aspect 3. The method of Aspect 2, wherein the generating of the convex combinations is based on combination coefficients generated by inputting global features of the extracted image features to a multi-layer perceptron (MLP).

(71) Aspect 4. The method of any of Aspects 1-3, wherein the extracted image features include image features extracted from input images.

(72) Aspect 5. A non-volatile computer-readable medium having computer-executable instructions stored thereon that, when executed, causes at least one processor to perform operations related to object detection comprising: receiving an input image; generating dynamic detection queries based on basic queries and semantics of the input image; training a transformer-based object detection model using the basic queries and the dynamic detection queries; and performing object detection based on the dynamic detection queries using the trained transformer-based object detection model.

(73) Aspect 6. The non-volatile computer-readable medium of Aspect 5, wherein the generating of the dynamic detection queries comprises: calculating a global feature as a global average pooling of a feature map of the semantics of the input image; generating combination coefficients by inputting the global feature to a multi-layer perceptron (MLP); and performing a convex calculation of the combination coefficients and the basic queries to generate the dynamic detection queries.

(74) Aspect 7. The non-volatile computer-readable medium of Aspect 6, wherein the calculating is based on features extracted from input images.

(75) Aspect 8. The non-volatile computer-readable medium of any of Aspects 5-7, wherein the performing of object detection using the trained transformer-based object detection model includes replacing the basic queries with the modulated queries.

(76) Aspect 9. The non-volatile computer-readable medium of any of Aspects 5-7, wherein the performing of object detection is by a DETR transformer decoder.

(77) Aspect 10. The non-volatile computer-readable medium of any of Aspects 5-9, wherein the at least one processor executes on a server corresponding to a social media platform.

(78) Aspect 11. The non-volatile computer-readable medium of any of Aspects 5-9, wherein the at least one processor executes on a smart device.

(79) Aspect 12. A transformer-based object detector, comprising: a feature extraction module; a global pooling module to calculate combination coefficients using output from the feature extraction module; a dynamic detection query generator to generate dynamic detection queries as a function of the combination coefficients and basic queries for a transformer-based object detection model; and a transformer decoder to receive the output from the feature extraction module and the

dynamic detection queries as input for performing object detection.

(80) Aspect 13. The transformer-based object detector of Aspect 12, wherein the detector executes a DETR object detection model.

(81) Aspect 14. The transformer-based object detector of either Aspect 12 or Aspect 13, wherein the global pooling module: receives the output from the feature extraction module; calculates a global feature of the extracted images; and inputs the global feature to a multi-layer perceptron (MLP) to generate the combination coefficients.

(82) Aspect 15. The transformer-based object detector of any of Aspects 12-14, wherein the output from the feature extraction module includes features extracted from input images.

(83) Aspect 16. The transformer-based object detector of any of Aspects 12-15, wherein the dynamic detection query generator generates the dynamic detection queries by performing a convex calculation of the combination coefficients and the basic queries.

(84) Aspect 17. The computer-readable medium of any of Aspects 12-16, wherein the input for performing object detection excludes the basic queries for a transformer-based feature extraction module.

Claims

1. A method of detecting objects using a transformer-based object detection model, comprising: generating modulated queries based on basic queries of a transformer-based object detection model and extracted image features of an input image; training the transformer-based object detection model using the basic queries and the modulated queries; replacing the basic queries with the modulated queries as input for a transformer decoder that is to execute the trained transformer-based object detection model; and inputting the modulated queries and the extracted image features to the transformer decoder.

2. The method of claim 1, wherein the generating of the modulated queries includes generating convex combinations of the basic queries to produce the modulated queries.

3. The method of claim 2, wherein the generating of the convex combinations is based on combination coefficients generated by inputting global features of the extracted image features to a multi-layer perceptron (MLP).

4. The method of claim 1, wherein the extracted image features include image features extracted from input images.

5. A non-volatile computer-readable medium that, when executed, causes at least one processor to perform operations related to object detection comprising: receiving an input image; generating dynamic detection queries based on basic queries and semantics of the input image; training a transformer-based object detection model using the basic queries and the dynamic detection queries; and performing object detection based on the dynamic detection queries using the trained transformer-based object detection model.

6. The non-volatile computer-readable medium of claim 5, wherein the generating of the dynamic detection queries comprises: calculating a global feature as a global average pooling of a feature map of the semantics of the input image; generating combination coefficients by inputting the global feature to a multi-layer perceptron (MLP); and performing a convex calculation of the combination coefficients and the basic queries to generate the dynamic detection queries.

7. The non-volatile computer-readable medium of claim 6, wherein the calculating is based on features extracted from input images.

8. The non-volatile computer-readable medium of claim 5, wherein the performing of object detection using the trained transformer-based object detection model includes replacing the basic queries with modulated queries.

9. The non-volatile computer-readable medium of claim 5, wherein the performing of the object detection is by a DETR transformer decoder.

10. The non-volatile computer-readable medium of claim 5, wherein the at least one processor executes on a server corresponding to a social media platform.
 11. The non-volatile computer-readable medium of claim 5, wherein the at least one processor executes upon a smart device.
 12. An apparatus comprising: at least one processor; and at least one computer-readable medium that, when executed, causes the at least one processor to perform operations comprising: extracting features from an input image; calculating combination coefficients using output from the features; generating dynamic detection queries as a function of the combination coefficients and basic queries for a transformer-based object detection model; and performing object detection based on the features and the dynamic detection queries.
 13. The apparatus of claim 12, wherein the apparatus executes a DETR object detection model.
 14. The apparatus of claim 12, wherein calculating the combination coefficients comprise: receiving the features extracted from the input image; calculating a global feature of the extracted images; and inputting the global feature to a multi-layer perceptron (MLP) to generate the combination coefficients.
 15. The apparatus of claim 14, wherein the dynamic detection queries are generated by performing a convex calculation of the combination coefficients and the basic queries.
 16. The apparatus of claim 12, wherein inputs for performing the object detection excludes the basic queries.
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